K. Kulachart Parnduangkeaw

Data Science & Machine Learning · Computer Engineering, SWU

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Key results **0.89 AUC-ROC** skin-cancer screening, 3,619 unseen images / **0.84 recall** peak-demand days vs 0.73 baseline / **Top 30** KBTG Hackathon 2026

Summary

Computer Engineering student at SWU (expected 2027) and Data Science Intern at Botnoi Group (2026). Builds ML end to end, from leakage-safe validation and threshold tuning to models that run in the browser, plus LLM-powered apps with Gemini. Seeking a Data Science / AI internship.

Experience

Data Science Intern — Botnoi Group · May – Jul 2026

- Built product demand forecasting for SME clients: data cleaning, feature engineering, model training and evaluation.
- Set up the MLOps pipeline to train and deploy the models, and built dashboards for the client companies.

Projects

Derma AI: Skin Cancer Screening — CPE310 Healthcare AI · Solo project · 2026

- Fine-tuned ImageNet-pretrained ResNet50, EfficientNet-B3 and DenseNet121 on ISIC 2019 (25,331 images, 8 classes) with class-weighted loss for the imbalanced classes.
- Split by lesion (StratifiedGroupKFold) so no test lesion is seen in training; tuned each model's cancer threshold on validation to target 90% sensitivity.
- Best model: **0.89 AUC-ROC** and 88.7% top-2 accuracy on 3,619 unseen test images; exported to ONNX so it runs in the browser and images never leave the device.

Python · PyTorch · scikit-learn · Pandas · Deep Learning · Computer Vision · ONNX Runtime · React

Peak Electricity Day Alert — CPE312 Data Science capstone · 2026

- Forecast a week ahead which days hit peak demand from 2015–2020 Panama load and weather data, with leakage-safe daily features and a time-based train/test split.
- XGBoost caught **0.84 of peak days vs 0.73** for a same-day-last-week baseline, at higher precision (0.78 vs 0.73); also compared Logistic Regression.

Python · XGBoost · scikit-learn · Pandas · Matplotlib · Machine Learning

Smart Flood AI — DOH Hackathon 2026 · Team of 3 · Web & AI dev · 2026

- Zero-shot Grounding DINO finds cars and wheels in highway CCTV to **estimate water depth** from how far the wheels are submerged, with no labeled training data.
- Fed a FastAPI + Next.js command center with flood-aware routing and LINE OA alerts.

Python · PyTorch · OpenCV · Grounding DINO · FastAPI · Next.js · Firebase · Computer Vision

KaTang Cash-flow Co-pilot — KBTG Hackathon 2026 · Team of 3 · Data science · 2026

- Built the gradient-boosting spending forecast and Monte Carlo prototype (simulated data) behind a safe daily spending limit for variable-income first jobbers; **pitched in the top-30 round**.

Python · scikit-learn · Pandas · Machine Learning · Next.js · TypeScript

WELLYNC — True Innovation 2026 · Team of 5 · Web & LLM dev · 2026

- Chronic-care prototype for patients, caregivers and doctors with multi-day early-warning alerts; built **Gemini tools** that read blood-pressure and glucose monitor screens, estimate meal nutrients, and draft care plans.

Next.js · TypeScript · Tailwind CSS · Generative AI · Gemini

Education

Bachelor of Engineering in Computer Engineering — Srinakharinwirot University (SWU) · 2024 – Present · Expected 2027
Relevant coursework: Artificial Intelligence, Healthcare AI Systems, Database Systems, Software Development
Google Cloud, Big Data & GenAI Training — Google Cloud, 2026

Leadership & Teaching

Computer Engineering Student Leader — Dept. of Computer Engineering, SWU · 2024 – Present

- Represent CPE students at faculty meetings; plan and run workshops and student-development programs.

Robotics Teaching Assistant — Srinakharinwirot University · 2026

- Taught robotics and programming fundamentals to high-school students in hands-on sessions.

Skills

ML / DL: PyTorch, scikit-learn, XGBoost, Hugging Face Transformers, OpenCV, ONNX Runtime, Gemini API
Methods: Demand forecasting, transfer learning, threshold tuning, time-based validation, Monte Carlo, MLOps
Data: Pandas, NumPy, Matplotlib, Seaborn, MySQL, Google Colab
Languages & web: Python, SQL, TypeScript, JavaScript, Java, C/C++, FastAPI, React, Next.js, Firebase, Google Cloud, Git